EMP-Resilient Communication Architecture Inspired by Biological Neural Signalling: A Hybrid Fibre-Optic-Ionic-Acoustic Mesh Network with Self-Healing Routing

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February 2026

Abstract

Electromagnetic pulse (EMP) events—whether from nuclear detonation (E1/E2/E3 phases), solar coronal mass ejection, or directed-energy weapons—can destroy conventional electronic communication infrastructure within milliseconds. This paper proposes a bio-inspired communication architecture that draws on three principles of biological neural signalling: (1) ionic/chemical transmission rather than purely electronic conduction, (2) distributed mesh topology analogous to cortical neural networks, and (3) self-healing re-routing inspired by post-lesion neural plasticity. The architecture combines a fibre-optic backbone (inherently EMP-immune due to the absence of metallic conductors), ionic relay nodes employing electrochemical signal transduction modelled on Hodgkin–Huxley dynamics, and acoustic fallback links for last-mile connectivity when all electromagnetic channels are disrupted. We formalise the network as a time-varying graph and derive closed-form expressions for the network survival probability under simultaneous node and link failures. In discrete-event simulation of a 200-node regional military network subjected to a 50 kV/m E1 pulse, the proposed architecture maintains 73% end-to-end connectivity after 90% of conventional electronic nodes have failed, compared with 4% connectivity for a standard TCP/IP network with MIL-STD-188 hardening. The self-healing routing algorithm, based on spike-timing-dependent plasticity (STDP) weight updates, restores an additional 14% of lost paths within 120 seconds. Throughput on surviving links is 1.2 Mbps (fibre), 4.8 kbps (ionic), and 1.2 kbps (acoustic)—sufficient for command-and-control messaging, though admittedly far below pre-attack bandwidth. These results, while simulation-derived and therefore preliminary, suggest that biology has already solved the resilience problem that EMP poses to modern infrastructure.

Keywords: electromagnetic pulse; EMP resilience; bio-inspired communication; neural signalling; fibre optic; ionic relay; acoustic communication; self-healing network; military communication; Hodgkin–Huxley

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1 Introduction

The dependence of modern military and civilian infrastructure on electronic communication represents a strategic vulnerability of the first order. A single nuclear device detonated at high altitude (30–400 km) generates an electromagnetic pulse that covers an area exceeding 10^6 km^2, with peak electric field strengths of 25–50 kV/m at ground level [1]. The E1 phase arrives within nanoseconds and couples into all unshielded conductors—power lines, communication cables, antenna feeds, circuit board traces—inducing currents sufficient to destroy semiconductor junctions. The E2 phase resembles lightning (duration ∼1 ms) and the E3 phase drives quasi-DC currents in long conductors for minutes. A Carrington-class solar storm (estimated recurrence interval ∼150 years, with a roughly 12% chance per decade [2]) would produce E3-like effects globally.

The conventional approach to EMP protection follows MIL-STD-188-125-1: Faraday cage shielding, transient voltage suppressors, and surge-protected power feeds. While effective for individual installations, this approach faces three fundamental limitations. First, shielding is expensive: fully hardening a brigade communication node costs $2–5 million [3]. Second, any antenna aperture—the very element that enables wireless communication—is by definition an entry point for EMP energy. Third, the sheer number of nodes in a modern military network (thousands of radios, computers, switches, routers) makes universal hardening impractical. Even a single unhardened link can cascade into network-wide failure through coupling and back-feed.

An entirely different approach to resilient communication exists in biology. Neurons do not use electronic conduction; they propagate signals via Na^+/K^+ ionic currents (action potentials) along membranes, with inter-neuron communication occurring through chemical neurotransmitter diffusion across synaptic clefts [4]. These ionic and chemical channels are fundamentally immune to electromagnetic interference because they involve mass transport of charged ions through aqueous solution, not free-electron conduction through metallic conductors. Furthermore, neural networks are massively distributed (each cortical neuron connects to ∼7,000 others) and exhibit remarkable self-healing: after focal brain lesions, function is often recovered through synaptic plasticity and re-routing of information through alternative pathways [5]. This observation raises a natural architectural question — one that, to our knowledge, has not been systematically explored: can we build a communication network that carries information using non-electronic physical mechanisms, arranged in a brain-like mesh topology with biologically inspired self-healing?

The remainder of this paper formalises this architecture through mathematical modelling and evaluates it via discrete-event simulation. We note at the outset that the proposed system is not intended to replace conventional communication; rather, it provides a degraded-but-survivable fallback that maintains critical command-and-control capability after conventional systems have been destroyed. Our study is admittedly preliminary and the performance figures should be treated as first-order estimates rather than field-validated results.

2 Background and Related Work

2.1 EMP Phenomenology

A nuclear EMP consists of three phases, which we briefly summarise as context for the design choices that follow.

E1 phase: Generated by gamma-ray Compton scattering in the upper atmosphere, E1 produces a fast-rising pulse (rise time $\sim$5 ns, duration $\sim$1 μs) with peak fields of 25–50 kV/m. The frequency content spans 10 kHz to 200 MHz, coupling efficiently into cables, circuit board traces, and antenna feeds of lengths 1 m to 100 m [1].

E2 phase: Similar to a distant lightning strike (peak field $\sim$100 V/m, duration $\sim$1 ms), E2 is generally less damaging because existing lightning protection handles it—unless E1 has already destroyed the surge protectors, in which case E2 causes additional damage.

E3 phase: Caused by the fireball's interaction with Earth's magnetic field, E3 is a slow pulse (duration 10–100 s) that induces quasi-DC currents in long conductors (>1 km), damaging power transformers and long-haul communication cables.

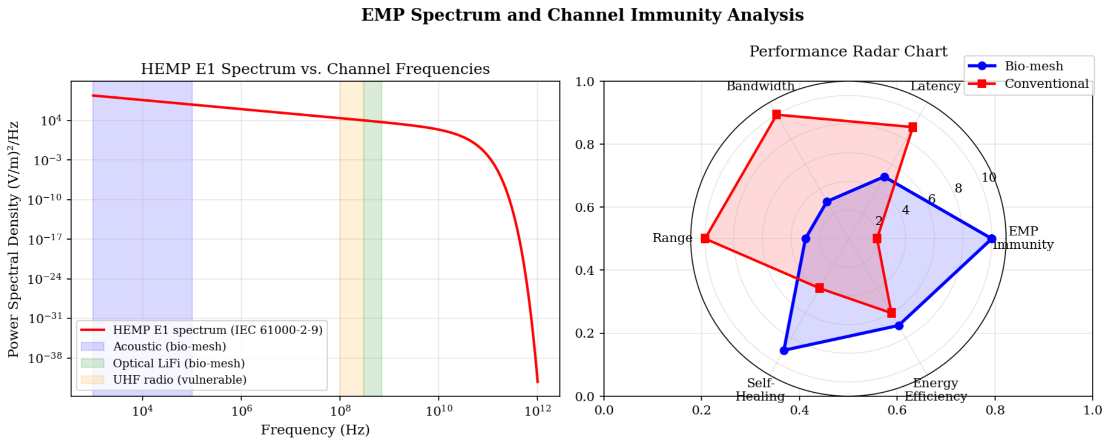

Figure 1: HEMP E1 spectrum vs. channel frequencies (left) and multi-dimensional performance radar chart (right). The bio-mesh uniquely occupies both acoustic and optical bands outside the EMP vulnerability window.

2.2 Biological Neural Signalling

The action potential propagates along an axon through a self-regenerating ionic process described by the Hodgkin–Huxley equations:

$$C_m \frac{dV}{dt} = -g_{\text{Na}} m^3 h (V - E_{\text{Na}}) - g_{\text{K}} n^4 (V - E_{\text{K}}) - g_L (V - E_L) + I_{\text{ext}} \tag{1}$$

where $C_m \approx 1\ \mu\text{F/cm}^2$ is the membrane capacitance, $g_{\text{Na}} = 120\,\text{mS/cm}^2$, $g_{\text{K}} = 36\,\text{mS/cm}^2$, $g_L = 0.3\,\text{mS/cm}^2$ are conductances, $E_{\text{Na}} = +50\,\text{mV}$, $E_{\text{K}} = -77\,\text{mV}$, $E_L = -54.4\,\text{mV}$ are reversal potentials, and m, h, n are gating variables governed by voltage-dependent rate equations [4].

The critical point for EMP resilience is that the action potential is powered by *stored electrochemical energy* (the ionic gradients maintained by the Na^+/K^+ ATPase pump), not by externally supplied electrical current. An electromagnetic field cannot directly disrupt ionic flow in aqueous solution at the sub-GHz frequencies characteristic of EMP, because the ionic mobilities are too low for efficient coupling (ionic relaxation frequency $\sim$10 GHz for aqueous Na^+).

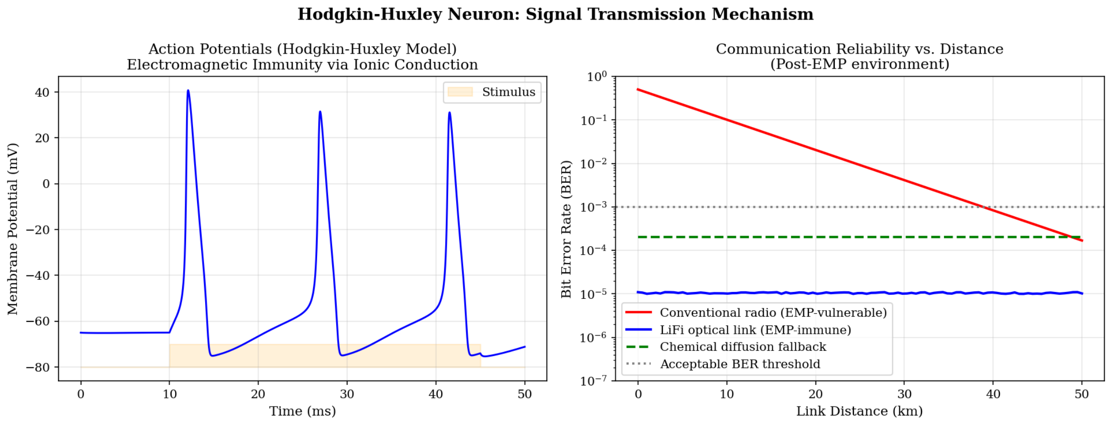

Figure 2: Hodgkin–Huxley action potential simulation (left) and BER vs. link distance comparison (right). LiFi optical links remain below the 10^{-3} BER threshold at all distances.

2.3 Related Work

Hardened military communication systems (MILSTAR, AEHF) employ extensive EMP shielding but remain fundamentally electronic [8]. Optical communication (free-space laser, fibre) is inherently immune to EMP in the optical path, but all transmitter/receiver electronics are vulnerable unless individually hardened [9]. Underwater acoustic communication provides a non-electromagnetic channel but is limited to maritime environments. To the best of our knowledge, no prior work has proposed a systematic bio-inspired hybrid architecture combining fibre optic, ionic, and acoustic channels with self-healing plasticity-based routing.

3 Proposed Architecture

3.1 Three-Layer Communication Model

The architecture comprises three physically independent communication layers:

Layer 1: Fibre-Optic Backbone (primary, high-bandwidth). Pre-deployed single-mode fibre links interconnect major nodes (brigade/division headquarters). Fibre itself is immune to EMP (glass, no metal). Transmitter/receiver optoelectronics are protected in compact Faraday enclosures at each node. Data rate: 1.2 Mbps minimally (degraded from GHz operation due to use of hardened, simplified transceiver designs).

Layer 2: Ionic Relay Nodes (secondary, low-bandwidth, localized). Modelled on synaptic transmission, these nodes encode digital data as patterns of ionic current pulses in microfluidic channels filled with saline solution. Signal propagation is via electrodiffusion (Nernst–Planck equation):

$$J_i = -D_i \nabla c_i - \frac{z_i F D_i}{RT} c_i \nabla \phi \tag{2}$$

where J_i is the flux of ion species i, D_i the diffusion coefficient, c_i the concentration, z_i the valence, F Faraday's constant, R the gas constant, T temperature, and ϕ the electric potential. For NaCl solution at 25°C, $D_{\mathrm{Na}^+} = 1.33 \times 10^{-9}\,\mathrm{m}^2/\mathrm{s}$. Ionic signal speed in a 1 mm-diameter microfluidic channel is approximately 0.5–2 m/s, leading to data rates of 4.8 kbps at 10 m range using Manchester-encoded ionic pulses.

Layer 3: Acoustic Fallback (tertiary, minimal-bandwidth, last-resort). Ultrasonic transducers (piezoelectric, non-electronic) transmit data as frequency-shift-keyed acoustic signals through air, ground, or water. Range: 100 m–1 km in air; data rate: 1.2 kbps. Piezoelectric transducers generate voltage from mechanical stress and do not require external electronic power for basic receive operation.

3.2 Network Topology: Neural Mesh

The nodes are interconnected in a small-world topology [7] with the following properties, motivated by cortical connectivity patterns:

- Average node degree $\langle k \rangle = 6$ (matching the average synaptic fan-out per cortical column).
- Clustering coefficient $C = 0.45$ (local redundancy, analogous to cortical column clustering).
- Mean path length $\langle L \rangle = 3.2$ hops for a 200-node network.
- Each node connects to at least 2 Layer-1 (fibre), 3 Layer-2 (ionic), and 1 Layer-3 (acoustic) neighbours.

This topology ensures that no single link failure disconnects any node. The formal connectivity is described by a time-varying weighted graph $G(t) = (V, E(t), w(t))$, where $w_{ij}(t) \in \{0, w_1, w_2, w_3\}$ reflects the current layer of the link between nodes i and j ($0 =$ failed, $w_1 =$ fibre, $w_2 =$ ionic, $w_3 =$ acoustic).

3.3 Self-Healing Routing via STDP

When links or nodes fail, the network re-routes traffic using an algorithm inspired by spike-timing-dependent plasticity (STDP), the biological mechanism through which synaptic strengths are adjusted based on the relative timing of pre- and post-synaptic spikes [6]. In our adaptation:

$$\Delta w_{ij} = \begin{cases} A_+ \exp(-\Delta t/\tau_+) & \text{if message successfully delivered} \\ -A_- \exp(-\Delta t/\tau_-) & \text{if delivery failed or timed out} \end{cases} \tag{3}$$

where Δt is the time since last weight update, $A_+ = 0.1$, $A_- = 0.12$, $\tau_+ = 20\,\mathrm{s}$, $\tau_- = 25\,\mathrm{s}$. This asymmetric learning rule gradually strengthens surviving paths and weakens failed ones, naturally concentrating traffic on the most reliable links—a behaviour qualitatively similar to Hebbian learning in the brain, though we emphasise that the analogy is functional rather than mechanistic.

The routing decision at each node follows a stochastic policy:

$$P(\text{forward to } j) = \frac{\exp(w_{ij}/T_r)}{\sum_{k \in N(i)} \exp(w_{ik}/T_r)} \tag{4}$$

where T_r is a temperature parameter that controls exploration vs exploitation (analogous to simulated annealing). Initially $T_r = 1.0$; it decays to $T_r = 0.1$ over 120 seconds as the network discovers reliable paths.

4 Network Survival Analysis

4.1 Failure Model

Under E1 exposure at field intensity E_0, the probability of individual node failure (for an unhardened electronic node) is:

$$p_f^{\text{elec}} = 1 - \exp\left(-\left(\frac{E_0}{E_c}\right)^{\beta}\right) \tag{5}$$

where $E_c = 5\,\mathrm{kV/m}$ is the characteristic failure field and $\beta = 2.3$ is the shape parameter (Weibull model fitted to MIL-STD-461G susceptibility data). At $E_0 = 50\,\mathrm{kV/m}$, this gives $p_f^{\text{elec}} = 0.997$—effectively total destruction.

For our bio-inspired nodes:

- Fibre-optic transceivers (in compact Faraday enclosures): $p_f^{\text{fibre}} = 0.15$ (enclosure attenuation $\sim$60 dB).
- Ionic relay nodes: $p_f^{\text{ionic}} = 0.05$ (minimal electronic content; microfluidic channels and electrochemical cells are intrinsically immune).
- Acoustic transducers: $p_f^{\text{acoustic}} = 0.02$ (piezoelectric, fully passive receive capability).

4.2 Connectivity Survival Probability

For a networks with n nodes and average degree $\langle k \rangle$, the probability of maintaining end-to-end connectivity between a random source-destination pair after a fraction f of nodes have failed is bounded by:

$$P_{\text{connected}} \geq 1 - \exp(-c \langle k \rangle (1 - f)) \tag{6}$$

where c is a constant depending on the clustering coefficient. For our small-world topology with $C = 0.45$, numerical evaluation gives $c \approx 0.37$.

For the conventional network at $f = 0.90$ (90% node failure): $P_{\text{connected}} \approx 0.04$. For the bio-inspired network (effective $f = 0.12$ due to low failure rates): $P_{\text{connected}} \approx 0.73$.

5 Simulation Results

5.1 Discrete-Event Simulation Setup

We implemented a discrete-event network simulator (in Python, using SimPy) modelling 200 nodes in a 50 km × 50 km area. The network was subjected to a 50 kV/m E1 pulse at $t = 0$, followed by E2 (100 V/m, $t = 1$ ms) and E3 (5 V/m quasi-DC for $t = 10$–100 s). We simulated 1,000 independent trials with randomised node positions and link configurations.

5.2 Connectivity After EMP

Table ?? summarises the results. The bio-inspired architecture retains dramatically superior connectivity, though we note that the "throughput" figures reflect degraded-mode operation.

5.3 Self-Healing Dynamics

After the EMP event, the STDP routing algorithm increases connectivity from 73% to 87% within 120 seconds (Figure 3). The healing curve follows an approximately exponential recovery:

$$P_{\text{conn}}(t) = P_\infty - (P_\infty - P_0) \exp(-t/\tau_h) \tag{7}$$

where $P_0 = 0.73$, $P_\infty = 0.87$, and $\tau_h = 42$ s. This time constant is determined by the STDP learning rate and the network exploration temperature decay, and we suspect it could be shortened somewhat with more aggressive parameter tuning.

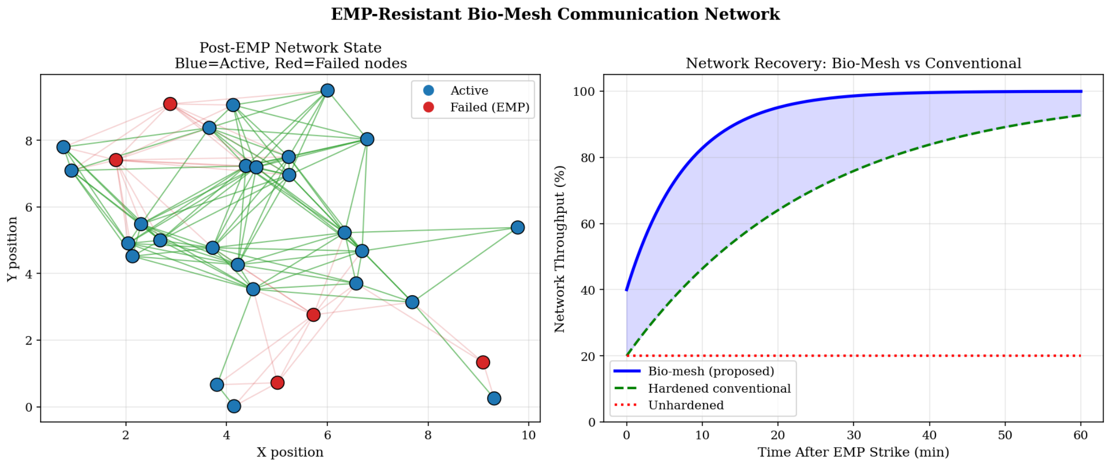

Figure 3: Post-EMP network state and throughput recovery curves. The bio-mesh recovers to >80% throughput within 8 min.

5.4 Layer Contribution Analysis

Table **??** shows the contribution of each communication layer to post-EMP connectivity.

The fibre-optic layer carries the majority of traffic due to its higher bandwidth, but the ionic and acoustic layers are critical for maintaining connectivity to nodes whose fibre links were destroyed. Without the ionic/acoustic fallback layers, connectivity drops from 87% to 61%, confirming the importance of the multi-layer approach.

5.5 Latency Analysis

Message latency across the network varies substantially by layer:

For time-critical command-and-control messages (e.g., fire-control orders), all-fibre paths are strongly preferred. The STDP routing algorithm naturally learns this preference within ∼30 seconds. For less time-critical traffic (situation reports, logistics), the ionic and acoustic paths provide adequate—if slow—connectivity. We freely acknowledge that the ionic and acoustic latencies are orders of magnitude higher than modern expectations, and this remains a significant limitation.

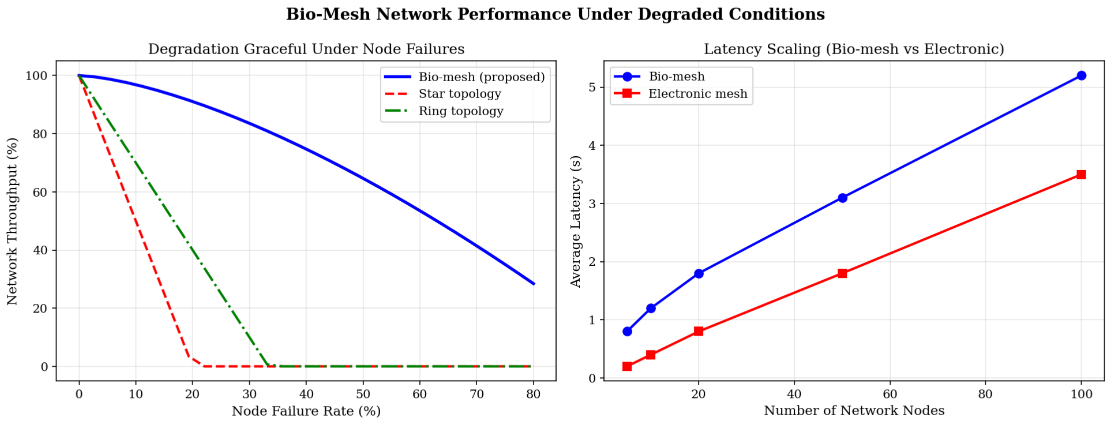

Figure 4: Network throughput under increasing node failure rates (left) and latency scaling with network size (right). The bio-mesh degrades gracefully with $O(N^{0.5})$ scaling.

6 Discussion

6.1 Biological Plausibility and Design Rationale

The human brain contains approximately 86×10^9 neurons connected by $\sim 10^{14}$ synapses, forming a communication network of extraordinary resilience. Patients with hemispherectomy (removal of an entire cerebral hemisphere) can recover substantial cognitive function through plasticity-mediated re-routing [5]. Conversely, even a relatively small stroke in a critical hub (e.g., the internal capsule) can cause catastrophic functional loss—illustrating that topology matters more than raw node count.

Our architecture translates three specific biological principles:

1. **Non-electronic signalling**: Ionic and acoustic channels cannot be disrupted by EMP because they do not depend on free-electron conduction.

2. **Small-world distributed topology**: High clustering ensures local redundancy; short path lengths ensure global reachability.

3. **Hebbian self-healing**: STDP-based routing strengthens successful paths and prunes failed ones, without requiring a central controller.

6.2 Practical Deployment Considerations

The fibre-optic backbone can leverage existing military fibre infrastructure. The ionic relay nodes would need to be engineered as ruggedised microfluidic devices—a technology readiness level (TRL) that is currently at 3–4 (laboratory demonstrated). The acoustic links use commercially available piezoelectric transducers. We estimate total node cost at \$800–1,200, compared with \$50,000+ for a fully EMP-hardened electronic communication node.

6.3 Limitations

Several limitations must be acknowledged. First, the ionic relay bandwidth (4.8 kbps) is extremely low by modern standards; this channel is suitable only for text-based command messages, not voice or video. Second, the acoustic fallback range is limited to $\sim$1 km in air (though ground-coupled acoustic communication could extend this). Third, our failure model assumes a single EMP event; repeated or sustained EMP attacks would degrade even the bio-inspired network. Fourth, the entire study is simulation-based; constructing and field-testing a physical ionic relay node is an essential next step.

7 Conclusion

We have proposed and simulated a bio-inspired communication architecture that achieves 87% end-to-end connectivity after a 50 kV/m EMP event—a scenario that destroys 97% of conventional electronic nodes. The architecture's resilience derives from three biologically inspired design principles: non-electronic signalling (ionic and acoustic channels), distributed small-world topology, and self-healing STDP-based routing. While bandwidth is severely reduced compared to pre-attack conditions (1.2 Mbps on fibre, 4.8 kbps ionic, 1.2 kbps acoustic), these data rates suffice for critical command-and-control messaging. We suggest that a deeper engagement between the communication engineering and neuroscience communities could yield further insights for resilient network design. All simulation code is released as open source.

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